

Three Frameworks for Designing and Analyzing Mathematical Assessments

Math Assessment Design Framework

These three frameworks are designed to be distinct, complementary lenses for designing and analyzing math assessments, moving far beyond simplistic "easy/medium/hard" labels. Each highlights a unique focus, justification, and distinguishing features.

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1 Skill Progression: Mapping the Learning Journey

Core Purpose: To assess where a student is along the continuum of mastering and autonomously applying a specific mathematical skill or concept.

Focus: The student's developmental path in handling a particular mathematical idea.

Level	Focus	Student Action	Justification Beyond Difficulty
Foundational	Grasping the core components & mechanics.	Executing basic steps, recalling definitions.	Ensures prerequisite mastery; focuses on accuracy and fluency with the raw skill.
Applied	Using the skill independently in standard situations.	Solving routine problems directly using the skill.	Assesses transfer to predictable contexts; measures functional understanding.
Strategic	Integrating & adapting the skill within complex demands.	Selecting/combining skills, devising multi-step approaches, justifying choices.	Measures true conceptual capacity and problem-solving for novel challenges.

Why it's distinct & justified: This framework is about **mastery trajectory**. A "Strategic" problem requires a qualitatively different kind of thinking about the *same* core concept—synthesis and judgment. It asks: *"Can the student use this math effectively when it's not handed to them on a plate?"*

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2 Cognitive Depth: Demanding the Type of Thinking

Core Purpose: To assess the complexity of mental processes required, regardless of the specific skill's difficulty.

Focus: The intellectual demand placed on the student's thinking.

gray!10 Level	Focus	Student Action	Justification Beyond Difficulty
Recall/Compute	Accessing stored knowledge or performing algorithmic steps.	Remembering facts, formulas; executing calculations.	Essential baseline. Assesses working memory and basic procedural knowledge.
Analyze/Interpret	Making sense of relationships, structures, or representations.	Comparing, contrasting, inferring patterns, translating representations.	Moves beyond rote application to seeing <i>how</i> and <i>why</i> math works.
Evaluate/Create	Making judgments or generating novel solutions.	Critiquing arguments, designing solutions, formulating models, proving conjectures.	Fosters higher-order thinking and creativity; uses math as a tool for innovation.

Why it's distinct & justified: This framework isolates **cognitive load**. A complex calculation might still only be "Recall/Compute," while a simple but insightful comparison is "Analyze/Interpret." It asks: *"What kind of thinking does this task require?"*

3 Real-World Relevance: Connecting Math to Context

Core Purpose: To assess the ability to connect mathematical abstractions to practical situations and interpret/use math meaningfully within them.

Focus: The nature and complexity of the context surrounding the math.

gray!10 Level	Focus	Nature of Context	Justification Beyond Difficulty
Procedural	Math as an isolated system.	Bare numbers, symbols, generic "word problems" with obvious math hooks.	Builds comfort with pure mathematical structures and symbolic fluency.
Contextual	Math as a tool within a simplified scenario.	Situations hinting at real-world use, requiring interpretation into math.	Develops ability to see math <i>in</i> the world; assesses translation and initial modeling.
Scenario-Based	Math as an integrated part of a complex situation.	Authentic scenarios requiring problem definition, assumptions, and interpretation.	Bridges the gap between classroom and life; assesses critical modeling and justification.

Why it's distinct & justified: This is about **contextualization**. Moving to "Scenario-Based" makes the interpretation and modeling choices the central challenge, even if the underlying math is simple. It asks: *"Can the student use math to make sense of a situation?"*

4 Key Distinctions & Holistic Assessment

Different Axes

Think of these as three separate dimensions:

- **Skill Progression:** How mastered is this specific skill? (*Novice* → *Expert*)
- **Cognitive Depth:** How complex is the thinking? (*Basic* → *Critical/Creative*)
- **Real-World Relevance:** How authentic is the context? (*Abstract* → *Authentic*)

Beyond "Difficulty"

A problem can exist at different intersections:

- **High Skill / Low Depth:** Executing a complex, multi-step algorithm from memory.
- **High Depth / Low Skill:** Critiquing the logic of a simple, foundational proof.
- **High Relevance / Moderate Skill:** Interpreting real-world dataset trends using standard techniques.

Justification

These frameworks align with established educational goals: **Skill Progression** (Mastery Learning), **Cognitive Depth** (Bloom's Taxonomy/DOK), and **Real-World Relevance** (Mathematical Modeling/Situated Learning).